Rabiat Sadiq

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EDUCATION

Carnegie Mellon University Pittsburgh, PA
M.S. in Computer Science, Human Computer Interaction, concentration in AI/ML and XR. *August 2026*
The University of Texas at San Antonio San Antonio, TX
B.S. in Computer Engineering, Minor in Computer Science. *December 2024*

EXPERIENCE

Technical Product Manager, Design Technologist and UX Researcher January 2026 – August 2026
Amazon Music, Carnegie Mellon MHCI Capstone

- Led mixed-methods research on music discovery and fan belonging: 74-response survey (61 variables), 8 semi-structured interviews spanning casual listeners to established artists, 10+ field intercepts, and record-store ethnography mapping tactile browsing behaviors and physical playback rituals.
- Synthesized 600+ affinity notes into a touchpoint/tension model and customer-lifecycle segmentation; ran 3 rounds of HMW reframing to narrow from feature ideation to system-level design principles.
- De-risked 5 design directions against a comprehension-by-desirability matrix; ran a Wizard-of-Oz study (5 moderated sessions) showing listeners will trade personal context for relevance when the system explains its reasoning and offers exit controls.
- Built and shipped the team's cross-platform prototype in TypeScript on React Native/Expo: 31 routes and 46 components from a single codebase targeting iOS, Android, and web; owned release management and Git workflow.
- Reverse-engineered Amazon Music's information architecture, layout, motion design, and native typography so the experimental adaptive-UI layer was the only isolated variable under test.

Graduate XR Researcher September 2025 – August 2026
Carnegie Mellon University, Augmented Perception Lab (XR) / HCII *Pittsburgh, PA*

- Build multimodal XR prototypes in Unity, investigating how visual, spatial audio, and ambient cues affect pain perception in controlled VR experiments.
- Design adaptive interfaces and log time-series user study data to support ML-driven feedback systems.
- Collaborate with interdisciplinary HCI faculty to advance human-centered XR and AI integration.

Applied Machine Learning Intern, Applied AI Team June 2025 – August 2025
PlayStation, Sony Interactive Entertainment *Aliso Viejo, CA*

- Built an end-to-end ML gameplay video context pipeline to extract on-screen scores and align outputs to timestamps for validation and downstream streaming analytics.
- Implemented parallel cloud ingestion from AWS S3 using multithreaded downloads and Ray task parallelism, reducing large video download time by 60%+.
- Automated preprocessing by splitting long-form streams into clips with FFmpeg, processing frames with OpenCV/OCR, and running scalable jobs in Databricks with PySpark.
- Tracked runs and artifacts with MLflow; improved data quality through iterative validation and performance-focused debugging.

AI/ML Program Participant June 2024 – August 2024
Apple, AI/ML Program *Los Angeles, CA*

- Built and deployed models across image classification, clustering, NLP, and responsible AI in an intensive 8-week applied ML program using Python, PyTorch, TensorFlow, CoreML, and scikit-learn.
- Engineered ML solutions using CNNs, transfer learning, SVMs, and decision trees; evaluated models with accuracy, precision, recall, F1 score, and ROC-AUC.
- Applied Swift and Xcode for on-device ML inference; leveraged linear algebra, statistical modeling, and feature engineering throughout hands-on projects.

AI/VR Undergraduate Research Assistant August 2023 – August 2024
UTSA *San Antonio, TX*

- Researched multimodal AI moderation for social VR safety, building Unity/C# data collection tooling and developing supervised and LLM-based behavior classification to reduce harassment and unsafe interactions.

Software Engineer Participant, Software Product Sprint Program Summer 2022 – Summer 2022
Google *Remote*

- Built Talky Talky, an audio-responsive web application for non-verbal children, using Java, JavaScript, HTML, CSS, and Google Text-to-Speech APIs with Google Cloud Platform services.

PROJECTS

Xbox x CMU: Measuring Social in Gaming January 2026 – August 2026
AI Research Engineer

- Partner with Xbox on an AI research project to measure and understand social interactions in gaming environments using AI-driven approaches.
- Develop methods to quantify social dynamics and player behavior to inform intelligent, human-centered gaming systems.

Magic Mitts: Affordable, Immersive VR Gloves January 2024 – December 2024

Lead Project Manager and Computer Engineer

- Invented a smart, programmable VR glove using a novel electromagnetic braking system and haptic actuators to deliver precise tactile feedback, simulating touch and grasp across standard VR systems.
- Developed algorithms for sensor processing, haptic control, and feedback synchronization; integrated flex-sensitive resistors for finger-tracking accuracy and a self-developed hardware driver for wireless, rechargeable operation.
- Placed 1st out of 89 teams at the UTSA Tech Symposium.

Sensing + ML Distress Prediction

- Built an end-to-end iOS sensing-to-insight pipeline transforming Apple HealthKit biometrics (HR, HRV, respiratory rate, steps, active energy, BMI) into real-time distress predictions.
- Trained and evaluated a logistic regression model achieving 82% test accuracy using pseudo-data augmentation; integrated on-device CoreML inference in Swift for live in-app feedback.

Applied STEM (Startup)

Co-founder, AI and Full-Stack

- Built a context-aware circuit interview mentor (React/TypeScript + FastAPI) that grounds LLM feedback in live circuit canvas state and simulation outputs to deliver adaptive questions and progressive hints.

SKILLS

ML and Computer Vision: Python, PyTorch, TensorFlow, scikit-learn, OpenCV, CoreML, CNNs, transfer learning, RNNs, embeddings, clustering, PCA, SVM, Computer Vision, EDA, model evaluation, feature engineering, cross-validation, Matplotlib/Seaborn, Image Processing, Object Detection

XR Engineering: Unity (C#), Meta XR SDK, VR prototyping, interaction systems, debugging

Video and Data Pipelines: FFmpeg, Ray, Databricks, PySpark, MLflow, SQL, Snowflake, multithreading, concurrency, Kafka

Engineering and Tooling: TypeScript, React Native, Expo, React, Swift, Xcode, Docker, Git/GitHub, CI/CD, Jupyter Notebooks, Jira, Confluence

Research Methods: Semi-structured interviews, contextual inquiry, participatory audits, ethnographic field research, survey design and analysis, affinity diagramming, Wizard-of-Oz prototyping, usability testing

Product and Design: Roadmapping and scoping, prioritization frameworks, stakeholder communication, release management, Figma, design systems, information architecture, rapid prototyping

Hardware: Microcontrollers, sensors and signal processing, haptics, electromagnetics, circuit design and analysis, embedded systems, Verilog, MATLAB

ACTIVITIES & LEADERSHIP

Professional Organizations

- IEEE, SWE, NACME, NSBE, RAAS (Robotics Association at UTSA)

AWARDS & HONORS

- Apple Scholar Program (January 2022)
- Horatio Alger Association Texas Scholarship (February 2021 and June 2023)
- Michael and Susan Dell Scholarship (March 2021)
- Marge and Bill Kleese ECE Scholarship (2022-2023)